SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 03:29 UTC

SEC P vs NP — daily log 2026-05-18

2 cycles recorded

Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete magnetic / signed Laplacian spectral theory of finite graphs — the $\mathbb{Z}/2$ Bochner–Laplacian L^σ for a line bundle on G representing a cohomology class in $H^1(G; F_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{min}(L^\sigma)$ (Lieb–Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai–Yates 2006; Atay–Liu 2014 and Lange–Liu–Peyerimhoff–Post 2015 on signed Cheeger inequalities). An arXiv search for ‘magnetic Laplacian’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the F_2 Bochner spectrum). × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and $\log_2$ t* controls V^0-witnessing complexity.
- **Recorded**: 2026-05-18 00:33 UTC
- **Entry ID**: `2d264de23a0e`

Statement

For a connected 3-regular graph $G=(V,E)$ with odd charge $\omega:V\to F_2$ ($\Sigma_v\ \omega(v)\equiv 1 \bmod 2$), build a $\mathbb{Z}/2$-gauge representative $\psi\in F_2^E$ by BFS from a fixed root v_0, setting each tree edge so that $\oplus_{e\ni v}\psi(e)=\omega(v)$ at every non-root v (the unsatisfiability deficit lands at v_0); form the signed adjacency A^σ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A}\sigma$, and define $\mu(G,\omega):=\lambda_{min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff $\Sigma\omega$ is even. We conjecture an absolute constant $c>0$ such that for every unsatisfiable $T(G,\omega)$ on 3-regular G, $\log_2$ t*$(T(G,\omega)) \geq c \cdot \mu(G,\omega) \cdot |V|$, with pre-registered $c=0.05$. A single 3-regular instance (G,ω) with $\log_2$ t*$(T(G,\omega)) < 0.05 \cdot \mu(G,\omega) \cdot |V|$ falsifies the conjecture.

Rationale

The Tseitin obstruction is exactly the $\mathbb{Z}/2$ holonomy class $[\omega]\in H^1(G;F_2)$, and $\mu(G,\omega)$ is the spectral norm of this obstruction in the discrete Bochner–Laplacian sense — large ($\Theta(1)$) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^2)$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson–Wigderson expansion-based $\exp(\Omega(\text{boundary expansion}))$ lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\geq\sqrt{2d\mu}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 158, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
       ^^^
NameError: name 'adj' is not defined
```

Judge reasoning

Test crashed due to undefined variable ‘adj’ in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the ‘adj’ variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction

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Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bonami-Beckner hypercontractivity (Bonami 1970, Beckner 1975) — the degree-d 4-norm-vs-2-norm contraction ||P||_4 ≤ (q-1)^{d/2}||P||_2 — applied as a SYNTACTIC Fourier-analytic invariant of the centered clause-indicator polynomial Q_F of a CNF (a property of the formula’s clause/polarity structure, NOT of its truth table). An arXiv search ‘Bonami-Beckner hypercontractivity’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’) returns 0 direct hits, with <5 adjacent papers (Schoenebeck-style PC-degree, never DPLL search-tree depth via syntactic clause-sum 4-2 ratios). Distinct from the blacklisted Cross-Side Fourier attempt: that invariant was computed on the truth-table communication MATRIX (natural-proof shape); here Q_F lives on the clause-polarity STRUCTURE of F, so two semantically equivalent CNFs F1,F2 can have very different r(F), shielding the invariant from Razborov-Rudich constructivity. × DPLL / tree-like Resolution refutation depth d*(F) for unsatisfiable 3-CNFs, in the Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where log$_2$ d*(F) controls V^0-witnessing complexity of ¬F as Σ^b_0 and lower-bounds tree-Resolution width w*(F).
- **Recorded**: 2026-05-18 03:29 UTC
- **Entry ID**: `2b2653395609`

Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, let s_{i,v} ∈ {±1} be the polarity sign of variable v in clause C_i (s=+1 if v occurs negated, -1 if positive). Define the centered clause-falsification polynomial Q_F: {-1,+1}^n → R by Q_F(x) = Σ_{i=1}^m (∏{v ∈ vars(C_i)} (1 + s{i,v}x_v)/2 − 1/8), whose Fourier coefficient at T ⊆ [n] with |T| ∈ {1,2,3} is the closed-form sum Q̂F(T) = (1/8) Σ{i: T ⊆ vars(C_i)} ∏{v ∈ T} s{i,v} (computable directly from clause structure in O(m) time). Let r(F) := ||Q_F||_4 / ||Q_F||_2 ∈ [1, 3^{3/2}] be the Bonami-Beckner 4-2 ratio, and set G(F) := (3^{3/2} − r(F)) / 3^{3/2} ∈ [0, 1−3^{-3/2}]. Conjecture: there is an absolute constant c ≥ 0.05 such that for every unsatisfiable 3-CNF F, $\log_2$(d*(F) + 1) ≥ c · n · G(F); a single F violating this inequality falsifies.

Rationale

Hypercontractive saturation r(F) → 1 means Q_F has a heavy-tailed degree-3 spectrum: clause-violation behaves like a sharp witness function whose Fourier mass spreads thin, the regime characterizing expander-Tseitin and threshold-random 3-SAT where DPLL must explore exponentially many literal interactions. Conversely r(F) → 3^{3/2} (Khintchine-saturated) means Q_F is dominated by a few large coefficients, the regime of structurally simple unsat formulas that DPLL refutes shallowly. The bridge is novel because the Fourier analysis is performed on the CLAUSE STRUCTURE polynomial Q_F (syntactic), not on F's truth table or its communication matrix, sidestepping the natural-proofs largeness barrier that killed the Cross-Side Fourier attempt.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.18s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 141, in run_trial
    norm_4 = compute_norm_4(F, n)
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 81, in compute_norm_
    product *= (1 + s * x[v - 1]) / 2
                        ~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in compute_norm_4 before producing any data, so the pre-registered support condition across 480 instances could not be evaluated. | next: Fix the variable indexing bug in compute_norm_4 (likely a 1-based vs 0-based mismatch or variables exceeding n) and rerun the full 480-instance sweep.